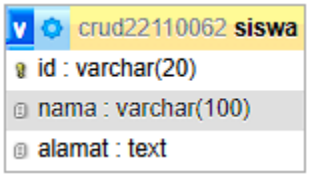
Nama : Pahni Hanawan
Nim : 22110062
Kelas : L 2022
MK : PBO

Modul 12

Database crud22110062 & Tabel siswa



File KoneksiMysql

```
1   package crudpbo;
2
3   import java.sql.Connection;
4   import java.sql.DriverManager;
5   import java.sql.SQLException;
6
7   public class KoneksiMysql {
8       private Connection connect;
9       private String driverName = "com.mysql.jdbc.Driver"; // Driver untuk koneksi ke MySQL
10      private String jdbc = "jdbc:mysql://";
11      private String host = "localhost"; // Bisa juga diganti IP seperti 192.168.100.100
12      private String port = "3306"; // Port default MySQL
13      private String database = "crud22110062"; // Nama database yang digunakan
14      private String url = jdbc + host + ":" + port + "/" + database;
15      private String username = "root"; // Username default MySQL
16      private String password = ""; // Kosongkan jika tidak ada password
17
18      public Connection getKoneksi() throws SQLException {
19          if (connect == null) {
20              try {
21                  Class.forName(driverName);
22                  System.out.println("Class Driver Ditemukan.");
23
24                  try {
25                      connect = DriverManager.getConnection(url, username, password);
26                      System.out.println("Koneksi Database Sukses.");
27                  } catch (SQLException se) {
28                      System.out.println("Koneksi Database Gagal: " + se);
29                      System.exit(0);
30                  }
31
32              } catch (ClassNotFoundException cnfe) {
33                  System.out.println("Class Driver Tidak Ditemukan, Terjadi Kesalahan Pada: " + cnfe);
34                  System.exit(0);
35              }
36          }
37
38          return connect;
39      }
40  }
41
```

File Crud

```java
package crudpbo;

import java.sql.*;

public class Crud extends SiswaAbstract implements SiswaInterface {
    Connection CRUDkoneksi;
    PreparedStatement CRUDpsmt;
    ResultSet CRUDrs;

    public Crud() {
        try {
            CRUDkoneksi = DriverManager.getConnection("jdbc:mysql://localhost/crud22110062", "root", "");
        } catch (SQLException e) {
            System.out.println("Koneksi gagal: " + e.getMessage());
        }
    }

    @Override
    public String simpanData(String id, String nama, String alamat) throws SQLException {
        try {
            String sql = "INSERT INTO siswa (id, nama, alamat) VALUES (?, ?, ?)";
            CRUDpsmt = CRUDkoneksi.prepareStatement(sql);
            CRUDpsmt.setString(1, id);
            CRUDpsmt.setString(2, nama);
            CRUDpsmt.setString(3, alamat);
            CRUDpsmt.executeUpdate();
            return "Data berhasil disimpan";
        } catch (SQLException e) {
            return "Data gagal disimpan: " + e.getMessage();
        }
    }

    @Override
    public ResultSet tampilData() throws SQLException {
        String sql = "SELECT * FROM siswa";
        CRUDpsmt = CRUDkoneksi.prepareStatement(sql);
        return CRUDpsmt.executeQuery();
    }

    @Override
    public String ubahData(String id, String nama, String alamat) throws SQLException {
        try {
            String sql = "UPDATE siswa SET nama=?, alamat=? WHERE id=?";
            CRUDpsmt = CRUDkoneksi.prepareStatement(sql);
            CRUDpsmt.setString(1, nama);
            CRUDpsmt.setString(2, alamat);
            CRUDpsmt.setString(3, id);
            CRUDpsmt.executeUpdate();
            return "Data berhasil diubah";
        } catch (SQLException e) {
            return "Data gagal diubah: " + e.getMessage();
        }
    }

    @Override
    public String hapusData(String id) throws SQLException {
        try {
            String sql = "DELETE FROM siswa WHERE id=?";
            CRUDpsmt = CRUDkoneksi.prepareStatement(sql);
            CRUDpsmt.setString(1, id);
            CRUDpsmt.executeUpdate();
            return "Data berhasil dihapus";
        } catch (SQLException e) {
            return "Data gagal dihapus: " + e.getMessage();
        }
    }

    @Override
    public ResultSet cariData(String keyword) throws SQLException {
        String sql = "SELECT * FROM siswa WHERE id LIKE ? OR nama LIKE ?";
        CRUDpsmt = CRUDkoneksi.prepareStatement(sql);
        CRUDpsmt.setString(1, "%" + keyword + "%");
        CRUDpsmt.setString(2, "%" + keyword + "%");
        return CRUDpsmt.executeQuery();
    }
}
```

File SiswaAbstract

```java
package crudpbo;

public abstract class SiswaAbstract {
    protected String id, nama, alamat;

    public void setID(String id) {
        this.id = id;
    }

    public void setNama(String nama) {
        this.nama = nama;
    }

    public void setAlamat(String alamat) {
        this.alamat = alamat;
    }

    public String getID() {
        return id;
    }

    public String getNama() {
        return nama;
    }

    public String getAlamat() {
        return alamat;
    }
}
```

File Siswa Interface

```java
package crudpbo;

import java.sql.ResultSet;
import java.sql.SQLException;

public interface SiswaInterface {
    public String simpanData(String id, String nama, String alamat) throws SQLException;
    public ResultSet tampilData() throws SQLException;
    public String ubahData(String id, String nama, String alamat) throws SQLException;
    public String hapusData(String id) throws SQLException;
    public ResultSet cariData(String keyword) throws SQLException;
}
```

File Main

```java
package crudpbo;

public class Main {
    public static void main(String[] args) {
        try {
            Form_Siswa form = new Form_Siswa();
            form.setVisible(true);
        } catch (Exception ex) {
            System.out.println(ex.toString());
        }
    }
}

```

File Form_Siswa

```java
package crudpbo;

import java.sql.ResultSet;
import java.sql.SQLException;
import javax.swing.JOptionPane;
import javax.swing.table.DefaultTableModel;

public class Form_Siswa extends javax.swing.JFrame {
 private DefaultTableModel tabmode;
    private ResultSet hasil;

    // Membuat objek dari class CRUD yang terhubung ke database
    Crud aa = new Crud();

    public Form_Siswa() throws SQLException {
        initComponents();
        tampil_database();
    }

    // Method untuk menampilkan data ke tabel siswa
    public void tampil_database() {
        Object[] baris = {"ID", "Nama", "Alamat"};
        tabmode = new DefaultTableModel(null, baris);
        table_siswa.setModel(tabmode);
        try {
            hasil = aa.tampilData();
            while (hasil.next()) {
                aa.setID(hasil.getString("id"));
                aa.setNama(hasil.getString("nama"));
                aa.setAlamat(hasil.getString("alamat"));
                String[] data = {aa.getID(), aa.getNama(), aa.getAlamat()};
                tabmode.addRow(data);
            }
        } catch (Exception e) {
//            System.out.println(e);
        }
    }
    public void tampil_cari(String keyword) {
    Object[] baris = {"ID", "Nama", "Alamat"};
    tabmode = new DefaultTableModel(null, baris);
    table_siswa.setModel(tabmode);
    try {
        hasil = aa.cariData(keyword);
        while (hasil.next()) {
            String[] data = {
                hasil.getString("id"),
                hasil.getString("nama"),
                hasil.getString("alamat")
            };
            tabmode.addRow(data);
        }
    } catch (Exception e) {
        JOptionPane.showMessageDialog(null, "Terjadi kesalahan saat mencari data");
    }
}
```

```
    // Method untuk reset input field
    public void reset_text() {
        txt_id.setText("");
        txt_nama.setText("");
        txt_alamat.setText("");
        txt_cari.setText("");
    }
    @SuppressWarnings("unchecked")
    Generated Code

    private void txt_namaActionPerformed(java.awt.event.ActionEvent evt) {
        // TODO add your handling code here:
    }

    private void btn_simpanActionPerformed(java.awt.event.ActionEvent evt) {
         if (txt_id.getText().trim().equals("")) {
            JOptionPane.showMessageDialog(null, "Maaf, ID belum diisi !", "Peringatan", JOptionPane.INFORMATION_MESSAGE);
            txt_id.requestFocus();
        } else if (txt_nama.getText().trim().equals("")) {
            JOptionPane.showMessageDialog(null, "Maaf, Nama belum diisi !", "Peringatan", JOptionPane.INFORMATION_MESSAGE);
            txt_nama.requestFocus();
        } else if (txt_alamat.getText().trim().equals("")) {
            JOptionPane.showMessageDialog(null, "Maaf, Alamat belum diisi !", "Peringatan", JOptionPane.INFORMATION_MESSAGE);
            txt_alamat.requestFocus();
        } else {
            try {
                aa.setID(txt_id.getText());
                aa.setNama(txt_nama.getText());
                aa.setAlamat(txt_alamat.getText());
                aa.simpanData(aa.getID(), aa.getNama(), aa.getAlamat());

                JOptionPane.showMessageDialog(null, "Data berhasil tersimpan", "Informasi", JOptionPane.INFORMATION_MESSAGE);
                tampil_database();
                reset_text();
            } catch (Exception e) {
                JOptionPane.showMessageDialog(null, "Data gagal tersimpan", "Informasi", JOptionPane.INFORMATION_MESSAGE);
                System.out.println(e);
            }
        }
    }

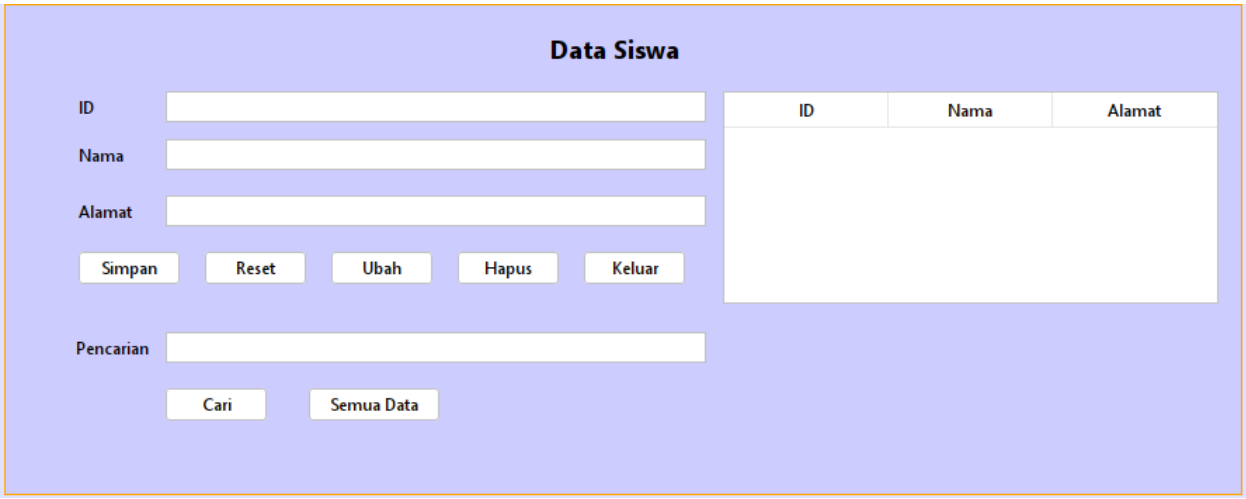
    private void btn_resetActionPerformed(java.awt.event.ActionEvent evt) {
       reset_text();
    }

    private void btn_ubahActionPerformed(java.awt.event.ActionEvent evt) {
         if (txt_id.getText().trim().equals("")) {
        JOptionPane.showMessageDialog(null, "Maaf, ID belum diisi !", "Peringatan", JOptionPane.INFORMATION_MESSAGE);
        txt_id.requestFocus();
    } else if (txt_nama.getText().trim().equals("")) {
        JOptionPane.showMessageDialog(null, "Maaf, Nama belum diisi !", "Peringatan", JOptionPane.INFORMATION_MESSAGE);
        txt_nama.requestFocus();
    } else if (txt_alamat.getText().trim().equals("")) {
        JOptionPane.showMessageDialog(null, "Maaf, Alamat belum diisi !", "Peringatan", JOptionPane.INFORMATION_MESSAGE);
        txt_alamat.requestFocus();
    } else {
        try {
            aa.setID(txt_id.getText());
            aa.setNama(txt_nama.getText());
            aa.setAlamat(txt_alamat.getText());
            aa.ubahData(aa.getID(), aa.getNama(), aa.getAlamat());

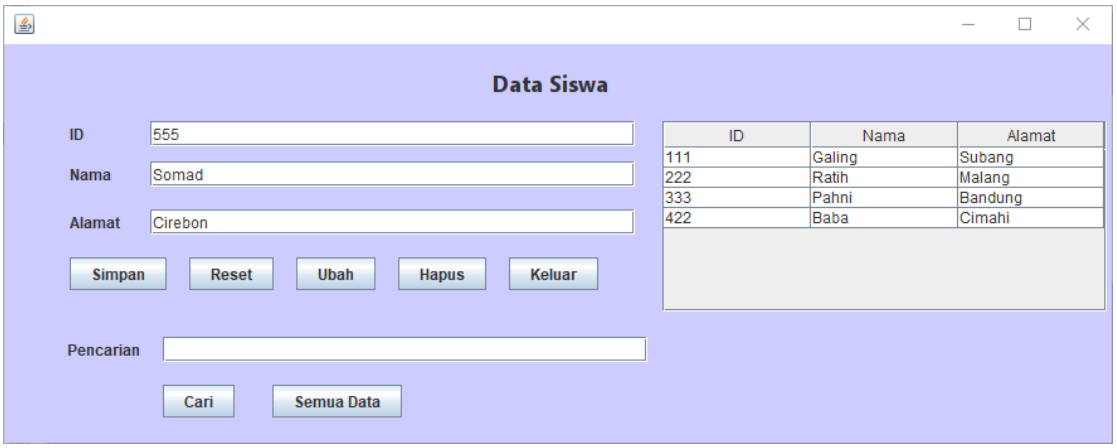
            JOptionPane.showMessageDialog(null, "Data berhasil diubah", "Informasi", JOptionPane.INFORMATION_MESSAGE);
            tampil_database();
```

```
341             } catch (Exception e) {
342                 JOptionPane.showMessageDialog(null, "Data gagal diubah", "Informasi", JOptionPane.INFORMATION_MESSAGE);
343             }
344         }
345     }
346 
347     private void btn_hapusActionPerformed(java.awt.event.ActionEvent evt) {
348         if (txt_id.getText().trim().equals("")) {
349             JOptionPane.showMessageDialog(null, "Maaf, ID belum diisi !", "Peringatan", JOptionPane.INFORMATION_MESSAGE);
350             txt_id.requestFocus();
351         } else {
352             int konfirmasi = JOptionPane.showConfirmDialog(null, "Apakah Anda yakin akan menghapus data ini?", "Warning", JOptionPane.YES_NO_OPTIO
353             if (konfirmasi == JOptionPane.YES_OPTION) {
354                 try {
355                     aa.hapusData(txt_id.getText());
356                     JOptionPane.showMessageDialog(null, "Data berhasil dihapus", "Informasi", JOptionPane.INFORMATION_MESSAGE);
357                     tampil_database();
358                     reset_text();
359                 } catch (Exception e) {
360                     JOptionPane.showMessageDialog(null, "Data gagal dihapus", "Informasi", JOptionPane.INFORMATION_MESSAGE);
361                 }
362             }
363         }
364 
365     private void btn_keluarActionPerformed(java.awt.event.ActionEvent evt) {
366         int konfirmasi = JOptionPane.showConfirmDialog(null, "Apakah Anda yakin akan keluar?", "Warning", JOptionPane.YES_NO_OPTION);
367     if (konfirmasi == JOptionPane.YES_OPTION) {
368         System.exit(0);
369     }
370     }
371 
372     private void table_siswaMouseClicked(java.awt.event.MouseEvent evt) {
373         try {
374         int row = table_siswa.rowAtPoint(evt.getPoint());
375 
376         String id = table_siswa.getValueAt(row, 0).toString();
377         String nama = table_siswa.getValueAt(row, 1).toString();
378         String alamat = table_siswa.getValueAt(row, 2).toString();
379 
380         txt_id.setText(id);
381         txt_nama.setText(nama);
382         txt_alamat.setText(alamat);
383     } catch (Exception e) {
384         JOptionPane.showMessageDialog(null, "Terjadi kesalahan saat mengambil data dari tabel.");
385     }
386     }
387 
388     private void txt_idActionPerformed(java.awt.event.ActionEvent evt) {
389         // TODO add your handling code here:
390     }
391 
392     private void btn_cariActionPerformed(java.awt.event.ActionEvent evt) {
393         String keyword = txt_cari.getText().trim();
394     if (!keyword.isEmpty()) {
395         tampil_cari(keyword);
396     } else {
397         JOptionPane.showMessageDialog(null, "Masukkan ID atau Nama untuk mencari", "Peringatan", JOptionPane.WARNING_MESSAGE);
398     }
399     }
400 
401     private void btn_semuadataActionPerformed(java.awt.event.ActionEvent evt) {
402         tampil_database();  // Menampilkan semua data lagi
403     }
404 
405 
406     // Variables declaration - do not modify
407     private javax.swing.JButton btn_cari;
408     private javax.swing.JButton btn_hapus;
409     private javax.swing.JButton btn_keluar;
410     private javax.swing.JButton btn_reset;
411     private javax.swing.JButton btn_semuadata;
412     private javax.swing.JButton btn_simpan;
413     private javax.swing.JButton btn_ubah;
414     private javax.swing.JLabel jLabel1;
415     private javax.swing.JLabel jLabel2;
416     private javax.swing.JLabel jLabel3;
417     private javax.swing.JLabel jLabel4;
418     private javax.swing.JLabel jLabel5;
419     private javax.swing.JPanel jPanel1;
420     private javax.swing.JScrollPane jScrollPane1;
421     private javax.swing.JTable table_siswa;
422     private javax.swing.JTextField txt_alamat;
423     private javax.swing.JTextField txt_cari;
424     private javax.swing.JTextField txt_id;
425     private javax.swing.JTextField txt_nama;
426     // End of variables declaration
427 }
428 
```

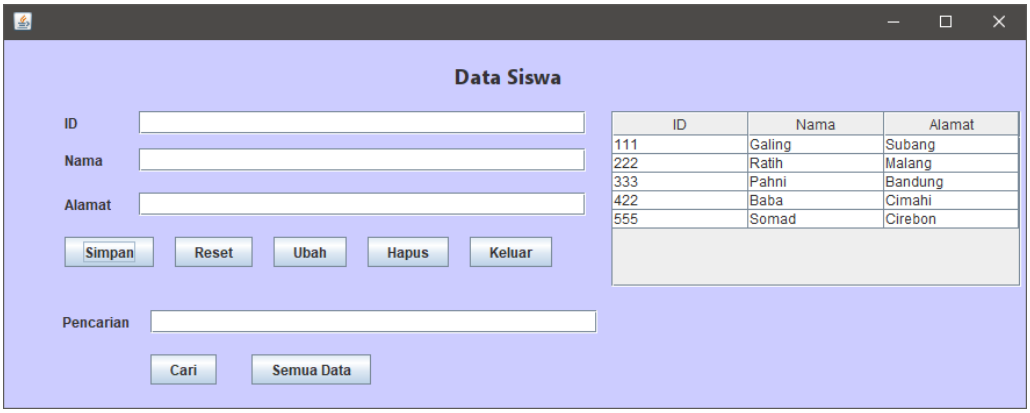
Desain



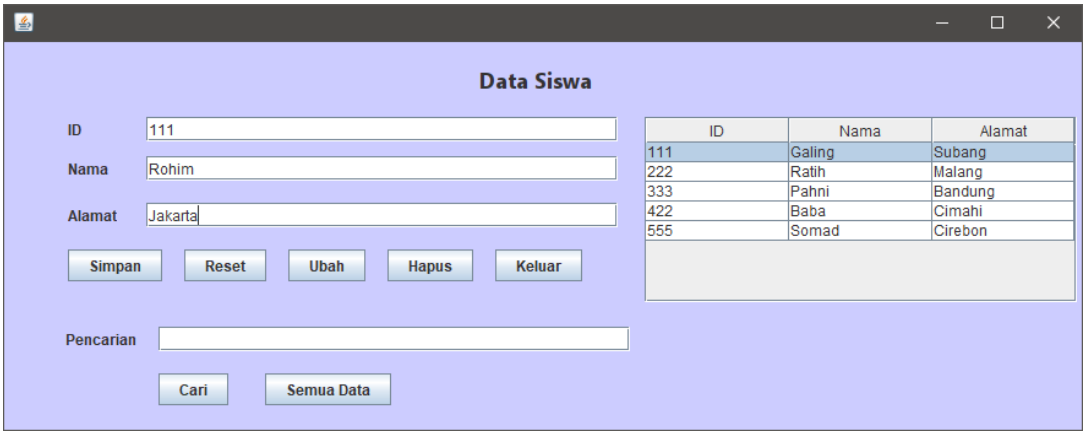
Tambah Data

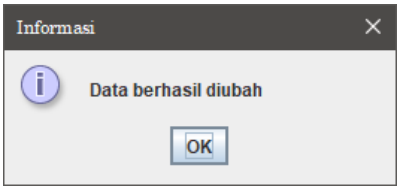


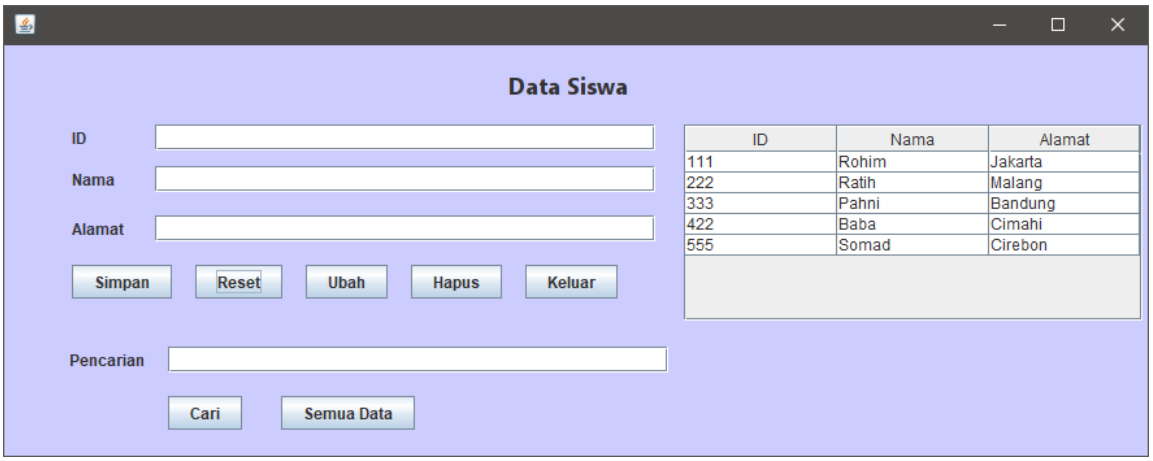




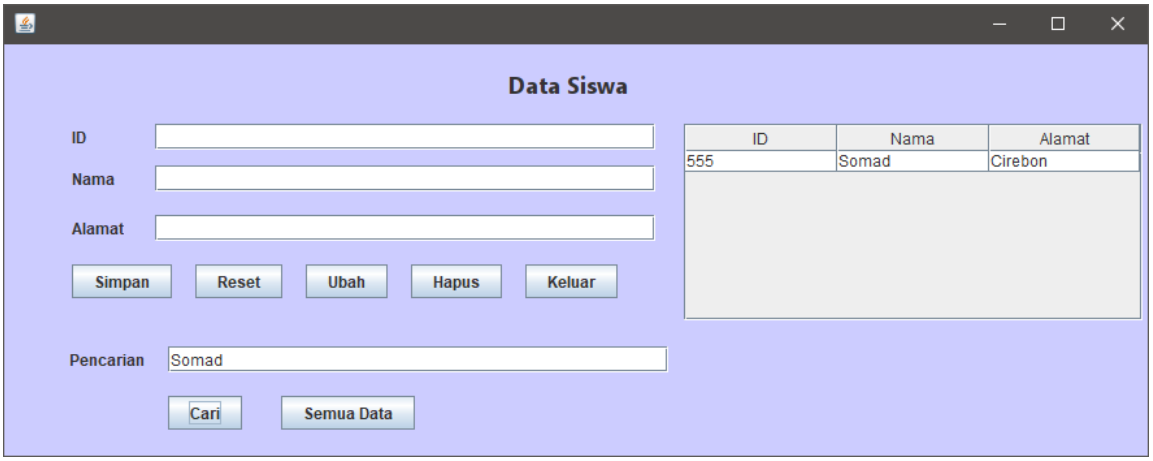
Ubah Data







Pencarian Data Berdasarakan Nama



Hapus Data (Menghapus Data Baba)

